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Inventor(s)

KUBOTA; HITOSHI

IMAGE FORMING APPARATUS WITH A STOPPER MEMBER AND AN URGING MEMBER

Abstract

An image forming apparatus includes an apparatus body, a photosensitive member unit, a developing unit, a pressing member movable between a first position for pressing the developing unit and a second position for releasing pressure, a stopper member movable between a regulating position for preventing the pressing member from moving from the second position to the first position and a retracting position for permitting the pressing member to move from the second position to the first position, and an urging member configured to urge the stopper member, wherein, in a case where the developing unit and the photosensitive member unit are attached to the apparatus body, the stopper member is at the retracting position, and wherein, in a case where the developing unit is attached to the apparatus body and the photosensitive member unit is not attached to the apparatus body, the stopper member is at the regulating position.

Inventors: KUBOTA; HITOSHI (Tokyo, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

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Background/Summary

BACKGROUND

Field

[0001] The present disclosure relates to copiers, printers, facsimiles, and other image forming apparatuses having developing units attachable to and detachable from the image forming apparatuses independently of photosensitive member units.

Description of the Related Art

[0002] U.S. Publication No. 20170285571 discusses an image forming apparatus including a photosensitive member unit having a photosensitive member, and a developing unit having a developer carrier for bearing and carrying a developer to develop an electrostatic latent image formed on the photosensitive member. The developing unit is attachable to and detachable from an apparatus body independently of the photosensitive member unit. The image forming apparatus has doors for accessing the photosensitive member unit and the developing unit. When a user or service person opens a door, application of the pressure to the developing unit is released, allowing the photosensitive member unit and the developing unit to be attached, detached, or replaced.

[0003] U.S. Publication No. 20160209804 discusses an image forming apparatus in which, in a state where a photosensitive member unit and a developing unit are attached to an apparatus body, the photosensitive member unit is positioned relative to the apparatus body, and the developing unit is positioned relative to the photosensitive member unit. More specifically, in a state where the photosensitive member unit and the developing unit are attached to the apparatus body, the developing unit is positioned relative to the photosensitive member unit when bearings at both longitudinal ends of the developer carrier come into contact with bearings at both longitudinal ends of the photosensitive member.

[0004] The image forming apparatus discussed in U.S. Publication No. 20160209804 has a configuration in which the photosensitive member unit serves as a reference for positioning the developing unit. A case in this configuration, where the door can be closed even in a state where the developing unit is attached to the apparatus body and the photosensitive member unit is not attached to the apparatus body, will be considered below. In this case, if the door is closed in a state where the photosensitive member unit is not attached to the apparatus body, the developing unit is pressed. Accordingly, the developing unit not having been positioned takes an unexpected orientation and may possibly collide with other units in the apparatus body by the vibration, for example, during transportation of the image forming apparatus.

SUMMARY

[0005] The present disclosure is directed to ensuring the orientation of a developing unit positioned relative to a photosensitive member unit attached to the apparatus body, in a case where the developing unit is attached to the apparatus body and the photosensitive member unit is not attached to the apparatus body.

[0006] According to some embodiments, an image forming apparatus includes an apparatus body, a photosensitive member unit including a photosensitive member on which an electrostatic latent image is formed, and configured to be attachable to and detachable from the apparatus body and positioned relative to the apparatus body, a developing unit including a developer carrier for bearing and carrying a developer for developing the electrostatic latent image formed on the photosensitive member, and configured to be attachable to and detachable from the apparatus body and positioned relative to the photosensitive member unit attached to the apparatus body, a pressure member configured to be movable between a first position for pressing the developing unit toward

the photosensitive member unit and a second position for releasing application of pressure to the developing unit, a stopper member configured to be movable between a regulating position for preventing the pressing member from moving from the second position to the first position and a retracting position for permitting the pressing member to move from the second position to the first position, the retracting position being retracted from the regulating position, and an urging member configured to urge the stopper member, wherein, in a case where the developing unit is attached to the apparatus body and the photosensitive member unit is also attached to the apparatus body, the stopper member urged by the urging member is at the retracting position, and wherein, in a case where the developing unit is attached to the apparatus body and the photosensitive member unit is not attached to the apparatus body, the stopper member urged by the urging member is at the regulating position.

[0007] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic view illustrating a configuration of an image forming apparatus according to a first exemplary embodiment.

[0009] FIG. 2 is a schematic view illustrating a configuration of a developing unit according to the first exemplary embodiment.

[0010] FIG. 3 is a cross-sectional view illustrating a configuration of the developing unit according to the first exemplary embodiment.

[0011] FIG. 4 is a schematic view illustrating a configuration of a photosensitive member unit according to the first exemplary embodiment.

[0012] FIG. 5 is another schematic view illustrating a configuration of the photosensitive member unit according to the first exemplary embodiment.

[0013] FIG. 6 is a cross-sectional view illustrating a configuration of the photosensitive member unit according to the first exemplary embodiment.

[0014] FIGS. 7A and 7B are schematic views illustrating configurations of a pressing mechanism and a stopper mechanism according to the first exemplary embodiment.

[0015] FIGS. 8A and 8B are schematic views illustrating configurations of the pressing mechanism and the stopper mechanism according to the first exemplary embodiment.

[0016] FIGS. 9A and 9B are schematic views illustrating configurations of the pressing mechanism and the stopper mechanism according to the first exemplary embodiment.

[0017] FIG. 10 is an external view illustrating configurations of a pressing mechanism and a door according to a second exemplary embodiment.

[0018] FIG. 11 is a schematic view illustrating a configuration of a stopper mechanism according to the second exemplary embodiment.

[0019] FIGS. 12A and 12B are schematic views illustrating configurations of the pressing mechanism and the door according to the second exemplary embodiment.

[0020] FIG. 13 is a schematic view illustrating a configuration of the stopper mechanism according to the second exemplary embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0021] Various exemplary embodiments, features, and aspects of the present disclosure will be described in detail below with reference to the accompanying drawings. The following exemplary embodiments do not limit the present disclosure within the scope of the appended claims. Not all of the combinations of the features described in a first exemplary embodiment are indispensable to the solutions for the present disclosure. The present disclosure can be embodied for printers, various

printing apparatuses, copiers, facsimiles, multifunction peripherals, and other diverse applications.

Configuration of Image Forming Apparatus

[0022] A configuration of an image forming apparatus according to a first exemplary embodiment will be described below with reference to the schematic view in FIG. 1.

[0023] As illustrated in FIG. 1, an image forming apparatus 1 is what is called an intermediate transfer tandem apparatus having an apparatus body 100 including image forming units UY, UM, UC, and UK for four different colors disposed to face an intermediate transfer belt 30. More specifically, the image forming apparatus 1 includes four different image forming units UY, UM, UC, and UK for yellow (Y), magenta (M), cyan (C), and black (K), respectively. However, the number of colors is not limited to four, and the order of the colors is not limited to this order.

[0024] Each of the image forming units UY, UM, UC, and UK includes a developing unit 10 and a photosensitive member unit 20. The image forming unit UY (UM to UK) includes a combination of the developing unit 10 and the photosensitive member unit 20 each of which is independently formed.

[0025] A recording material conveyance process of the image forming apparatus 1 will be described below. A recording material S is stacked and stored in a recording material storage (cassette) 70 and is fed to a conveyance path 40 in synchronization with an image forming timing. The recording material S fed from the recording material storage (cassette) 70 is subjected to skew correction and timing correction and then sent to a secondary transfer portion T2. The secondary transfer portion T2 is a nip portion formed by a secondary outer transfer roller 31 and a secondary inner transfer roller 32 facing each other. When a predetermined pressure and a predetermined electrostatic load bias are applied, a toner image is absorbed on the recording material S.

[0026] A process for conveying the recording material S to the secondary transfer portion T2 has been described above. A process for forming an image conveyed to the secondary transfer portion T2 at a similar timing will be described below. The image forming units UY to UK will be described below. Since the configurations of the image forming units UY to UK are basically the same except for the toner color, the image forming unit UY for yellow will be described below on behalf of the units UY to UK.

[0027] The image forming unit UY mainly includes the developing unit 10 having a developing sleeve 11 as a developer carrier for bearing and carrying a developer to develop an electrostatic latent image formed on a photosensitive drum 21, and the photosensitive member unit 20 having the photosensitive drum 21 and a charge roller 22. The developing unit 10 and the photosensitive member unit 20 will be described below.

[0028] The surface of the photosensitive drum 21 rotatably driven is uniformly pre-charged by the charge roller 22, and then an electrostatic latent image is formed by an exposure unit 60 driven based on an image information signal. The electrostatic latent image formed on the photosensitive drum 21 is developed with toner by the developing sleeve 11 to be a visible image.

[0029] When applied with a predetermined pressure and a predetermined electrostatic load bias by a primary transfer unit 33 disposed to face the image forming unit UY across the intermediate transfer belt 30, the toner image formed on the photosensitive drum 21 is primarily transferred onto the intermediate transfer belt 30. A toner supply unit (not illustrated) supplies a two-component developer (a mixture of a toner and a magnetic carrier) from a toner bottle 50 set in the image forming unit UY to the developing unit 10 through a supply path 51.

[0030] The intermediate transfer belt 30, an endless belt stretched by various rollers such as the secondary inner transfer roller 32, a driven roller 34, and a tension roller 35, is rotatably driven in the direction of the arrow R1 in FIG. 1. The secondary inner transfer roller 32 also serves as a drive roller for driving the intermediate transfer belt 30. The above-described image forming process is performed for four different colors in parallel by the image forming units UY to UK. This process for each color is performed at such a timing that the toner image is transferred on top of the toner image of another color coming from the upstream which has been primarily transferred on the

intermediate transfer belt **30**. As a result, a full-color toner image is eventually formed on the intermediate transfer belt **30** and then conveyed to the secondary transfer portion T2.

[0031] With the above-described conveyance process and image forming process, secondary transfer is performed with matched timing of the recording material S and the full-color toner image at the secondary transfer portion T2. Then, the recording material S is conveyed to the fixing unit **80**, and the toner image is melted and fixed to the recording material S with a predetermined pressure and a predetermined amount of heat. The recording material S with an image fixed thereon is discharged onto a discharge tray **90** by a discharge roller **41**.

[0032] Generally, durability lives of the developing unit **10** and the photosensitive member unit **20** are shorter than the durability life of other image forming process units (e.g., the fixing unit **80**) included in the image forming apparatus **1**. For this reason, the developing unit **10** and the photosensitive member unit **20** may need to be replaced. Accordingly, the apparatus body **100** is provided with insertion slots (not illustrated) for attaching and detaching the developing units **10** and the photosensitive member units **20** for the image forming units UY to UK on the front face (the near side in the insertion direction or the near side in FIG. **1**). An openable and closable door (not illustrated) is provided to cover the insertion slot for each color or cover the insertion slots for all colors.

[0033] Each of the developing units **10** and the photosensitive member units **20** is independently attached to the apparatus body **100** in consideration of a different durability life setting and the easiness of replacement during downtime.

[0034] The insertion direction (longitudinal direction) of the developing unit **10** and the photosensitive member unit **20** is the same as the rotational axis direction of the developing sleeve **11** and the rotational axis direction of the photosensitive drum **21**. The developing unit **10** and the photosensitive member unit **20** operate while receiving a transmitted drive force and supplied with electric power from the apparatus body **100**. To allow the drive force transmission and the electric power supply between the apparatus body **100** and the developing unit **10** and between the apparatus body **100** and the photosensitive member unit **20**, the developing unit **10** and the photosensitive member unit **20** are attached to the apparatus body **100** via coupling members (not illustrated).

Configuration of Developing Unit

[0035] The configuration of the developing unit **10** according to the first exemplary embodiment will be described below with reference to the schematic view in FIG. **2** and the cross-sectional view in FIG. **3**.

[0036] As illustrated in FIG. **2**, the developing unit **10** includes a developing container **110** and the developing sleeve **11**. In the developing container **110**, the developing sleeve **11** is rotatably supported by bearings **13a** and **13b** disposed at both ends on the near and the far sides in the insertion direction (the direction of the arrow X in FIG. **2**). As discussed below, the bearings **13a** and **13b** also have a function of members for forming a gap between the developing sleeve **11** and the photosensitive drum **21**. The developing unit **10** is swingably supported with respect to the apparatus body **100** via fitting holes **115a** and **115b**.

[0037] The developing container **110** stores a two-component developer (hereinafter simply referred to as a developer) containing a nonmagnetic toner and a magnetic carrier. The inside of the developing container **110** is partitioned into two chambers (a developing chamber **16a** and a stirring chamber **16b**) by a partition wall **15**, as illustrated in FIG. **3**. The developing chamber **16a** and the stirring chamber **16b** are connected at both longitudinal ends of the developing container **110**.

[0038] A conveyance screw **12a** is rotatably supported in the developing chamber **16a**, and a stirring screw **12b** is rotatably supported in the stirring chamber **16b**. The conveyance screw **12a** and the stirring screw **12b** are adjacently disposed in parallel so that their rotational axis directions coincide with the rotational axis direction of the developing sleeve **11**. When the conveyance screw

12a and the stirring screw **12b** are driven, the developer is circularly conveyed through the developing chamber **16a** and the stirring chamber **16b**. The developing sleeve **11** bears and carries the developer supplied from the developing chamber **16a**.

[0039] A regulation blade **14** is provided at a position near the surface of the developing sleeve **11** to form a thin layer of the developer on the surface of the developing sleeve **11**. The developing sleeve **11** supplies the developer to the photosensitive drum **21** at the developing position facing the photosensitive drum **21**. Thus, an electrostatic latent image formed on the photosensitive drum **21** is developed.

Configuration of Photosensitive Member Unit

[0040] A configuration of the photosensitive member unit **20** according to the first exemplary embodiment will be described below with reference to the schematic views in FIGS. **4** and **5** and the cross-sectional view in FIG. **6**.

[0041] As illustrated in FIGS. **4** and **5**, the photosensitive member unit **20** includes the photosensitive drum **21** and the charge roller **22** rotatably supported by a housing **24**. Bearings **25a** and **25b** disposed at both ends on the near and the far sides in the insertion direction (direction of the arrow **X** in FIGS. **4** and **5**) are provided coaxially with the rotational axis of the photosensitive drum **21**.

[0042] The housing **24** is disposed so that, when the photosensitive member unit **20** is attached to the apparatus body **100**, the position facing the developing sleeve **11** of the developing unit **10** is opened, and a part of the photosensitive drum **21** is exposed from the opening position.

[0043] As illustrated in FIG. **6**, the charge roller **22** uniformly charges the surface of the rotatably driven photosensitive drum **21**. The photosensitive drum **21** is connected with a driving source (not illustrated) of the apparatus body **100** and rotated by a driving force transmitted from the driving source. Meanwhile, the charge roller **22** is pressed onto the photosensitive drum **21** and rotatably driven by the photosensitive drum **21**.

[0044] After the photosensitive drum **21** is charged by the charge roller **22**, an electrostatic latent image is formed on the photosensitive drum **21** by the exposure unit **60** disposed in the apparatus body **100**. The electrostatic latent image formed on the photosensitive drum **21** is developed with a toner to be a visible toner image, and the toner image formed on the photosensitive drum **21** is transferred to the recording material **S**. Then, a drum cleaning blade **23** in contact with the photosensitive drum **21** by a predetermined pressure removes the transfer residual toner on the photosensitive drum **21**.

[0045] The bearings **25a** and **25b** rotatably supporting the photosensitive drum **21** are brought into contact with the bearings **13a** and **13b**, respectively, rotatably supporting the developing sleeve **11** when the photosensitive member unit **20** is attached to the apparatus body **100**. Accordingly, the bearings **25a** and **25b** also have a function of gap forming members for securing the gap between the photosensitive drum **21** and the developing sleeve **11** (hereinafter this gap is referred to as an SD gap).

[0046] In a state where the developing unit **10** and the photosensitive member unit **20** are attached to the apparatus body **100**, the developing unit **10** is urged toward the photosensitive member unit **20**. This brings the bearings **13a** and **13b** into contact with the bearings **25a** and **25b**, respectively, ensuring the positions of the developing sleeve **11** and the photosensitive drum **21**.

[0047] In a state where the bearings **13a** and **13b** are in contact with the bearings **25a** and **25b**, respectively, in such a configuration, the bearings **13a**, **13b**, **25a**, and **25b** may possibly be damaged by rubbing when the photosensitive member unit **20** is inserted into or removed from the apparatus body **100** at the time of replacement of the photosensitive member unit **20**. For this reason, before making the photosensitive member unit **20** attachable to and detachable from the apparatus body **100**, the developing unit **10** and the photosensitive member unit **20** need to be separated to bring the bearings **13a** and **13b** out of contact with the bearings **25a** and **25b**, respectively. After bringing the bearings **13a** and **13b** out of contact with the bearings **25a** and **25b**, respectively, a sufficient

gap can be ensured between the bearings **13a** and **25a** and between the bearings **13b** and **25b**. This prevents these bearings from being damaged when the photosensitive member unit **20** is inserted into or removed from the apparatus body **100**.

Configurations of Pressing Mechanism and Stopper Mechanism

[0048] Configurations of the pressing mechanism and the stopper mechanism according to the first exemplary embodiment will be described below with reference to the schematic views in FIGS.

7A, 7B, 8A, 8B, 9A, and 9B.

[0049] FIGS. **7A** and **7B** illustrate a state where the developing unit **10** is attached to the apparatus body **100** and the photosensitive member unit **20** is not attached to the apparatus body **100**. As illustrated in FIGS. **7A** and **7B**, a handle **150** is at a position (hereinafter referred to as a second position) where application of the pressure to the developing unit **10** is released, and the developing unit **10** is separated from the developing position.

[0050] FIGS. **8A** and **8B** illustrate a state where the developing unit **10** is attached to the apparatus body **100**, and the photosensitive member unit **20** is also attached to the apparatus body **100**. As illustrated in FIGS. **8A** and **8B**, the handle **150** is at the second position, and the developing unit **10** is separated from the developing position.

[0051] In FIGS. **9A** and **9B**, the photosensitive member unit **20** is attached to the apparatus body **100** and the photosensitive member unit **20** is also attached to the apparatus body **100**. As illustrated in FIGS. **9A** and **9B**, the handle **150** is at a position (hereinafter referred to as a first position) where the developing unit **10** is pressed toward the photosensitive member unit **20**.

[0052] As illustrated in FIG. **7A**, the handle **150** is rotatably provided around an axis **150a** parallel to the axis of the developing unit **10** relative to the apparatus body **100**. A pressing member **150c** urged by the pressing spring **150b** as an urging member, and a regulation unit **150d** for regulating the rotating action of the handle **150** are provided at the tip of the handle **150**.

[0053] The developing unit **10** is provided in the apparatus body **100**. A fitting hole **115a** and a swing axis **170a** on the near side in the insertion direction (in the direction of the arrow X in FIG. **7A**) are swingably supported around a swing axis (not illustrated) fitting into a fitting hole **115b** on the far side in the insertion direction (in the direction of the arrow X in FIG. **7A**).

[0054] A drum rail **106** is provided in the apparatus body **100** to allow the photosensitive member unit **20** to be easily attached at a predetermined position in the apparatus body **100**. The drum rail **106** is provided with a positioning member **106a** protruding from the guide surface of the drum rail **106** on the near side in the insertion direction (in the direction of the arrow X in FIG. **7A**).

[0055] As illustrated in FIGS. **7A** and **7B**, the drum rail **106** is provided with a stopper mechanism **160** for regulating the rotation of the handle **150**. The stopper mechanism **160** is provided with a stopper member **160a** and a spring **160b** as an urging member for urging the stopper member **160a**.

[0056] In the state illustrated in FIG. **7A**, the handle **150** is at the second position, and the developing unit **10** is at a position separated from the photosensitive member unit **20** by its weight.

[0057] As illustrated in FIG. **8A**, in a state where the developing unit **10** and the photosensitive member unit **20** are attached to the apparatus body **100**, the developing unit **10** is not positioned relative to the photosensitive member unit **20**. If a vibration is applied from outside to the apparatus body **100**, for example, during transportation of the image forming apparatus **1** in a state where the developing unit **10** is not positioned as described above, the vibration may also possibly be transmitted to the developing unit **10**.

[0058] Assume an example case where the user or service person closes a door (not illustrated) for accessing the photosensitive member unit **20** or the developing unit **10** in a state where the handle **150** is at the second position.

[0059] In this case, the developing unit **10** attached to the apparatus body **100** is not pressed by the handle **150** and separated from the photosensitive member unit **20** by its weight and therefore becomes unstable.

[0060] If the user or service person carelessly moves the image forming apparatus **1** in a state

where the developing unit **10** is not positioned relative to the photosensitive member unit **20** in this way, the developing unit **10** vibrates, possibly colliding with a nearby unit in the apparatus body **100**.

[0061] Accordingly, the door is often provided with a fool-proof shape that interferes with a part of the handle **150** if the user or service person closes the door in a state where the handle **150** is at the second position. This prevents the image forming apparatus **1** from being moved as much as possible while the developing unit **10** is not positioned relative to the photosensitive member unit **20**, in an unstable orientation.

[0062] However, it may be desirable to take into consideration a case where, in a state where the photosensitive member unit **20** is not attached to the apparatus body **100** as illustrated in FIGS. 7A and 7B, the action to rotate the handle **150** from the second position to the first position is permitted, and the user or service person closes the door.

[0063] Although the developing unit **10** is pressed by the handle **150**, the developing unit **10** is not positioned relative to the photosensitive member unit **20** since the photosensitive member unit **20** is not attached to the apparatus body **100**. Accordingly, in a state where the photosensitive member unit **20** is not attached to the apparatus body **100**, the developing unit **10** not positioned may take an unexpected orientation, possibly colliding with other units in the apparatus body **100**, for example, by the vibration during transportation of the image forming apparatus **1**.

[0064] According to the first exemplary embodiment, if the handle **150** is rotated in a state where the photosensitive member unit **20** is not attached to the apparatus body **100** as illustrated in FIG. 7B, the regulation unit **150d** comes into contact with the stopper member **160a**. This prevents the rotating action of the handle **150**.

[0065] This prevents the door from being closed in a state where the photosensitive member unit **20** is not attached to the apparatus body **100** and the developing unit **10** is pressed by the handle **150**. Accordingly, this also prevents the developing unit **10** not positioned from colliding with other units in the apparatus body **100**, for example, by the vibration during transportation of the image forming apparatus **1**.

[0066] As illustrated in FIG. 8A, the photosensitive member unit **20** is attached by being pressed toward the far side in the insertion direction (in the direction of the arrow X in FIG. 7A) of the apparatus body **100** from the insertion slot. The photosensitive member unit **20** can be slidably mounted on the drum rail **106** as a guide member having a guide surface extending in the insertion/removal direction (longitudinal direction) of the photosensitive member unit **20**.

[0067] When the photosensitive member unit **20** is inserted, the positioning member **106a** as an engagement member is elastically deformed by being pressed by the photosensitive member unit **20** and then retracted in the drum rail **106**. Meanwhile, the photosensitive member unit **20** is provided with a communication hole **20b** having an opening on the lower face.

[0068] When the photosensitive member unit **20** is inserted up to a predetermined position in the apparatus body **100**, the positioning member **106a** having been retracted in the drum rail **106** engages with the communication hole **20b** as an engaged portion, and the photosensitive member unit **20** is attached to the apparatus body **100**.

[0069] When the positioning member **106a** engages with the communication hole **20b**, the photosensitive member unit **20** is prevented from moving toward the near side in the insertion direction of the photosensitive member unit **20**, and the photosensitive member unit **20** is fixed not to protrude from the apparatus body **100**.

[0070] In the states illustrated in FIGS. 8A and 8B, the bearings **13a** and **13b** of the developing sleeve **11** are not in contact with the bearings **25a** and **25b** of the photosensitive drum **21**, respectively.

[0071] On the other hand, the communication hole **20b** having an opening on the lower face of the photosensitive member unit **20** penetrates through the photosensitive member unit **20**, and is provided with the other opening on the front face of the photosensitive member unit **20** on the near

side in the insertion direction. A release member **26** is provided in the communication hole **20b**.

[0072] In a state where the photosensitive member unit **20** is attached to the apparatus body **100**, one end of the release member **26** protrudes from the front face of the photosensitive member unit **20**, and the other end thereof comes into contact with the positioning member **106a** engaged with the communication hole **20b**. When the end on the front face of the photosensitive member unit **20** is downwardly pressed (in the direction of the arrow E in FIG. 8B), the other end of the release member **26** presses down the positioning member **106a**.

[0073] Thus, the engagement between the positioning member **106a** and the communication hole **20b** can be released. More specifically, attachment of the photosensitive member unit **20** to the apparatus body **100** is released, so that the photosensitive member unit **20** can be detached from the apparatus body **100**.

[0074] In a state where the photosensitive member unit **20** is attached to the apparatus body **100** as illustrated in FIG. 8B, the photosensitive member unit **20** comes into contact with the stopper member **160a** to press the stopper member **160a** toward the spring **160b**. Accordingly, the stopper member **160a** is urged toward the near side in the insertion direction not only by a coupling spring of a coupling member (not illustrated) but also by the spring **160b**. More specifically, when the photosensitive member unit **20** is attached to the apparatus body **100**, the stopper member **160a** is pressed, enabling the rotating action of the handle **150**.

[0075] As illustrated in FIGS. 9A and 9B, the user or service person rotates the handle **150**. When the handle **150** is rotated, the pressing member **150c** urged by the pressing spring **150b** presses the developing unit **10** around the swing axis **170a** to urge and position the photosensitive member unit **20**. In this case, the bearings **13a** and **13b** of the developing sleeve **11** are in contact with the bearings **25a** and **25b** of the photosensitive drum **21**, respectively, and the developing unit **10** is positioned with the SD gap ensured. As a result, the door is closed in a state where the developing unit **10** is suitably positioned.

[0076] As discussed above, in the image forming apparatus **1** according to the first exemplary embodiment, the developing unit **10** is urged and positioned relative to the photosensitive member unit **20** that is positioned relative to the apparatus body **100**. In a state where the photosensitive member unit **20** is not attached to the apparatus body **100** in the image forming apparatus **1**, the handle **150** pressing the developing unit **10** comes into contact with the stopper member **160a** to prevent the rotating action of the handle **150**.

[0077] This prevents the door from being closed in a state where the photosensitive member unit **20** is not attached to the apparatus body **100** and the developing unit **10** is pressed by the handle **150**. For this reason, the developing unit **10** not positioned can be prevented from colliding with other units in the apparatus body **100**, for example, by the vibration during transportation of the image forming apparatus **1**.

[0078] This means that, in the configuration of the first exemplary embodiment, the photosensitive member unit **20** is positioned relative to the apparatus body **100**, and the developing unit **10** is positioned relative to the photosensitive member unit **20**. In this configuration according to the first exemplary embodiment, the stopper mechanism **160** urges the stopper member **160a** so that the stopper member **160a** is at the retracting position in a state where the photosensitive member unit **20** is attached to the apparatus body **100** and the developing unit **10** is also attached to the apparatus body **100**. In this configuration according to the first exemplary embodiment, the stopper mechanism **160** urges the stopper member **160a** so that the stopper member **160a** is at the retracting position in a state where the photosensitive member unit **20** is attached to the apparatus body **100** and the developing unit **10** is not attached to the apparatus body **100**. In this configuration according to the first exemplary embodiment, the stopper mechanism **160** urges the stopper member **160a** so that the stopper member **160a** is at the regulating position in a state where the photosensitive member unit **20** is not attached to the apparatus body **100** and the developing unit **10** is attached to the apparatus body **100**.

[0079] According to the first exemplary embodiment, in a configuration in which the photosensitive member unit **20** serves as a reference for positioning the developing unit **10**, the orientation of the developing unit **10** can be ensured in a case where the developing unit **10** is attached to the apparatus body **100** and the photosensitive member unit **20** is not attached to the apparatus body **100**.

[0080] According to a second exemplary embodiment, a door is provided on each of the image forming units UY to UK to cover the insertion slot for each color. The developing unit **10** is separated from the photosensitive member unit **20** in association with the opening action of each door. A configuration in which the developing unit **10** is pressed onto the photosensitive member unit **20** in association with the closing action of the door will be described below.

[0081] The second exemplary embodiment will be described below on the premise that the developing unit **10** is configured to be attachable to and detachable from the apparatus body **100**. The attachment and detachment of the photosensitive member unit **20** to/from the apparatus body **100** according to the second exemplary embodiment are similar to the attachment and detachment according to the first exemplary embodiment, and detailed descriptions thereof will be omitted.

[0082] FIG. **10** is an outside drawing illustrating configurations of the pressing mechanism and the door according to the second exemplary embodiment.

[0083] FIG. **11** is schematic view illustrating a configuration of the stopper mechanism **160** according to the second exemplary embodiment. More specifically, FIG. **11** illustrates a joint between the door and the pressing mechanism, and a configuration of the stopper mechanism **160** for regulating the door opening/closing action.

[0084] FIGS. **12A** and **12B** are schematic views illustrating configurations of the pressing mechanism and the door according to the second exemplary embodiment. More specifically, FIGS. **12A** and **12B** illustrate the door opening/closing action and the state of the pressing mechanism.

[0085] FIG. **13** is a schematic view illustrating a configuration of the stopper mechanism **160** according to the second exemplary embodiment. More specifically, FIG. **13** illustrates a state of the stopper mechanism **160** based on a combination of a state where the developing unit **10** and the photosensitive member unit **20** are attached to the apparatus body **100** and a state where neither unit is attached to the apparatus body **100**.

[0086] As illustrated in FIG. **10**, the pressing mechanism **200** includes a separation link **102** and a developing rail **103**. The developing unit **10** can be slidably mounted on the developing rail **103** including a guide surface extending in the insertion direction (longitudinal direction). When the developing unit **10** is slid on the developing rail **103**, the developing unit **10** is guided into the apparatus body **100**.

[0087] Under the developing rail **103**, there is disposed the separation link **102** extending in the insertion direction (longitudinal direction) along the developing rail **103**. The separation link **102** is provided with pressure levers **104** and **105** at both ends in the insertion direction, i.e., the pressure lever **104** on the far side in the insertion direction (downstream side in the insertion direction) and the pressure lever **105** on the near side in the insertion direction (upstream side in the insertion direction upstream).

[0088] As illustrated in FIG. **11**, the pressure levers **104** and **105** include fitting members **104a** and **105a**, respectively, formed so that the amount of upward protrusion in the vertical direction increases toward a door **101** from the far side of the apparatus body **100**.

[0089] As illustrated in FIG. **10**, the developing rail **103** includes fit members **103a** and **103b** that can be engaged with the fitting members **104a** and **105a** of the pressure levers **104** and **105**, respectively.

[0090] The door **101** is fastened to the separation link **102** via a link member **101b**.

[0091] The second exemplary embodiment includes a stopper mechanism **160'** having a stopper member **160'a** for regulating the closing action of the door **101**, and a first urging spring **160'b** as an urging member for urging the stopper member **160'a**. The second exemplary embodiment

includes a rotary axis **160'c** for rotatably supporting the stopper mechanism **160'** relative to the apparatus body **100**, and a second urging spring **160'd** as an urging member for urging the stopper mechanism **160'** around the rotary axis **160'c**.

[0092] The action of the pressing mechanism for pressing the developing unit **10** in association with the opening/closing action of the door **101** will be described below with reference to FIGS. **12A** and **12B**.

[0093] The door **101** is disposed at an insertion slot (not illustrated) of the apparatus body **100** so that the door **101** can rotate around a rotary axis **101a**, between a closing position where the door **101** covers the upstream end of the developing unit **10** in the insertion direction and an opening position where the door **101** exposes the upstream end of the developing unit **10** in the insertion direction.

[0094] The separation link **102** moves toward the near side in the insertion direction (second position) when the door **101** is opened, and moves toward the far side in the insertion direction (first position) when the door **101** is closed.

[0095] In a state where the door **101** is open (open state) illustrated in FIG. **12A**, the fitting members **104a** and **105a** fit into the fit members **103a** and **103b**, respectively.

[0096] When the fitting members **104a** and **105a** fit into the fit members **103a** and **103b**, respectively, the developing rail **103** is downwardly moved in the vertical direction (toward the installation surface of the image forming apparatus **1**) in FIG. **12A** to come closer to the separation link **102** than in a state where fitting members **104a** and **105a** do not fit into the fit members **103a** and **103b**.

[0097] More specifically, when the door **101** is opened from the closed state, the link member **101b** moves toward the near side in the insertion direction (see the arrow F in FIG. **12A**). Accordingly, the separation link **102** also moves to the second position on the near side in the insertion direction. When the separation link **102** moves to the second position, the fit members **103a** and **103b** slide down along the fitting members **104a** and **105a**, and then the fitting members **104a** and **105a** fit into the fit members **103a** and **103b**, respectively.

[0098] Thus, the developing rail **103** moves downwardly in the vertical direction (the direction of the arrow D in FIG. **12A**), and the developing unit **10** and the photosensitive member unit **20** separate from each other. In this case, the bearings **13a** and **13b** of the developing sleeve **11** are not in contact with the bearings **25a** and **25b** of the photosensitive drum **21**, respectively.

[0099] On the other hand, in a state where the door **101** is closed (closed state) illustrated in FIG. **12B**, the fitting member **104a** (the fitting member **105a** omitted from FIG. **12A**) does not fit into the fit member **103a** (the fit member **103b** omitted from FIG. **12A**). When the fitting members **104a** and **105a** do not fit into the fit members **103a** and **103b**, respectively, the developing rail **103** is upwardly moved in the vertical direction (toward the side opposite to the installation surface side of the image forming apparatus **1**) so that the developing rail **103** is separated from the separation link **102**.

[0100] More specifically, when the door **101** is opened, the link member **101b** moves toward the far side in the insertion direction, and the separation link **102** also similarly moves to the first position on the far side in the insertion direction. In a state where the developing unit **10** and the photosensitive member unit **20** are attached to the apparatus body **100**, the developing unit **10** is pressed onto the photosensitive member unit **20**. In this case, the bearings **13a** and **13b** of the developing sleeve **11** come into contact with the bearings **25a** and **25b** of the photosensitive drum **21**, respectively, and the developing unit **10** is positioned with the SD gap ensured.

[0101] Thus, as the developing rail **103** is vertically moved in association with the opening/closing action of the door **101**, the developing unit **10** is pressed onto and separated from the photosensitive member unit **20**.

[0102] A state where the developing unit **10** is attached to the apparatus body **100** and the photosensitive member unit **20** is not attached to the apparatus body **100** will be considered below.

In this case, a case where the user or service person closes the door **101** needs to be considered.

[0103] In association with the closing action of the door **101**, the separation link **102** moves to the first position, and the developing unit **10** is pressed. However, since the photosensitive member unit **20** is not attached to the apparatus body **100**, the developing unit **10** is not positioned and may possibly become unstable.

[0104] If the user or service person carelessly moves the image forming apparatus **1** in a state where the developing unit **10** is not positioned relative to the photosensitive member unit **20** in this way, the developing unit **10** vibrates, possibly colliding with a nearby unit in the apparatus body **100**.

[0105] According to the second exemplary embodiment, in a state where the photosensitive member unit **20**, relative to which the developing unit **10** is positioned and urged, is not attached to the apparatus body **100** as illustrated in FIG. **11**, a regulation member **101d** of the door **101** comes into contact with the stopper member **160'** to regulate the closing action of the door **101**.

[0106] This prevents the door **101** from being closed in a state where the photosensitive member unit **20**, relative to which the developing unit **10** is positioned, is not attached to the apparatus body **100** and the developing unit **10** is pressed. This prevents the image forming apparatus **1** from being moved as much as possible in a state where the developing unit **10** not positioned relative to the photosensitive member unit **20** is in an unstable orientation.

[0107] In a state where the photosensitive member unit **20** is attached to the apparatus body **100**, the stopper member **160'a** is pressed against the spring **160'b**, like in the first exemplary embodiment. More specifically, when the photosensitive member unit **20** is attached to the apparatus body **100**, the photosensitive member unit **20** presses the stopper member **160'a**, allowing the closing action of the door **101**.

[0108] More specifically, the user or service person closes the door **101** in a state where the developing unit **10** and the photosensitive member unit **20** are attached to the apparatus body **100**. As a result, as illustrated in FIG. **12B**, the separation link **102** moves to the first position, and the developing unit **10** is pressed against the photosensitive member unit **20**.

[0109] In this case, the bearings **13a** and **13b** of the developing sleeve **11** come into contact with the bearings **25a** and **25b** of the photosensitive drum **21**, respectively, and the developing unit **10** is positioned with the SD gap ensured. As a result, the door **101** is closed in a state where the developing unit **10** is suitably positioned, making it possible to prevent the developing unit **10** from colliding with a nearby unit.

[0110] A state where the developing unit **10** is attachable to and detachable from the apparatus body **100**, and neither the developing unit **10** nor the photosensitive member unit **20** is attached to the apparatus body **100** will be described below.

[0111] The stopper mechanism **160'** according to the second exemplary embodiment is rotatable around the rotary axis **160'c**. Accordingly, in a state where the developing unit **10** is not attached to the apparatus body **100**, the stopper mechanism **160'** is rotated by the second urging spring **160'd** as illustrated FIG. **13**. As a result, the regulation member **101d** of the door **101** and the stopper member **160'a** do not come into contact with each other, and the action to close the door **101** is not regulated.

[0112] According to the above-described first exemplary embodiment, the stopper member **160a** is urged so that the stopper member **160a** is at the regulating position in a state where neither the photosensitive member unit **20** nor the developing unit **10** is attached to the apparatus body **100**. The regulating position refers to a position where the pressing member (the handle **150** according to the first exemplary embodiment) pressing the developing unit **10** is prevented from moving from the second position for releasing application of the pressure to the developing unit **10** to the first position for pressing the developing unit **10** toward the photosensitive member unit **20**.

[0113] According to the second exemplary embodiment, on the other hand, the stopper member **160'a** is urged so that the stopper member **160'a** moves from the regulating position to the

retracting position in a state where neither the photosensitive member unit **20** nor the developing unit **10** is attached to the apparatus body **100**. The retracting position refers to a position where the pressing member (the pressure lever **104** according to the second exemplary embodiment) pressing the developing unit **10** is permitted to move from the second position for releasing application of the pressure to the developing unit **10** to the first position for pressing the developing unit **10** toward the photosensitive member unit **20**.

[0114] For this reason, according to the second exemplary embodiment, the user or service person can close the door **101** even in a state where neither the photosensitive member unit **20** nor the developing unit **10** is attached to the apparatus body **100**. In a state where neither the developing unit **10** nor the photosensitive member unit **20** is attached to the apparatus body **100**, units do not collide with each other even if the image forming apparatus **1** is moved.

[0115] On the other hand, in a state where the developing unit **10** is attached to the apparatus body **100**, the stopper mechanism **160'** is pressed onto the developing unit **10** and rotates around the rotary axis **160'c** up to the position illustrated in FIG. **11**. The effect of the subsequent action of the stopper mechanism **160'** is as described above with reference to the first exemplary embodiment.

[0116] The second exemplary embodiment has been described above centering on a configuration in which the separation link **102** moves in association with the opening/closing action of the door **101**. However, the present exemplary embodiment is not limited to the example. Like the above-described first exemplary embodiment, when the door **101** is rotated by the opening/closing action of the door **101**, the pressing mechanism may move as the door **101** rotates.

[0117] The present disclosure is not limited to the above-described exemplary embodiments and can be modified in diverse ways (including organic combinations of these exemplary embodiments) based on the spirit and scope thereof. These modifications are not excluded from the scope of the present disclosure.

[0118] According to the above-described exemplary embodiments, the photosensitive member unit **20** is insertable into and removable from the apparatus body **100**, and the developing unit **10** is brought into and out of contact with the photosensitive member unit **20**. However, the present exemplary embodiment is not limited to the examples. For example, the developing unit **10** may be insertable into and removable from the apparatus body **100**, and the photosensitive member unit **20** may be brought into and out of contact with the developing unit **10**.

[0119] According to the above-described exemplary embodiments, the image forming apparatus **1** primarily transfers the toner image carried by the photosensitive drum **21** onto the intermediate transfer belt **30**, as illustrated in FIG. **1**. Then, the image forming apparatus **1** brings the recording material **S** into contact with the intermediate transfer belt **30**, and secondarily transfers the toner image carried by the intermediate transfer belt **30** to the recording material **S**. Although the exemplary embodiments have been described above centering on the image forming apparatus **1** having such a configuration, the present disclosure is not limited to the examples. The present disclosure is also applicable to an image forming apparatus configured to directly bring the recording material **S** into contact with the photosensitive drum **21** and directly transfer the toner image carried by the photosensitive drum **21** to the recording material **S**.

[0120] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0121] This application claims the benefit of priority from Japanese Patent Application No. 2024-025974, filed Feb. 22, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image forming apparatus comprising: an apparatus body; a photosensitive member unit including a photosensitive member on which an electrostatic latent image is formed, and configured to be attachable to and detachable from the apparatus body and positioned relative to the apparatus body; a developing unit including a developer carrier for bearing and carrying a developer for developing the electrostatic latent image formed on the photosensitive member, and configured to be attachable to and detachable from the apparatus body and positioned relative to the photosensitive member unit attached to the apparatus body; a pressing member configured to be movable between a first position for pressing the developing unit toward the photosensitive member unit and a second position for releasing application of pressure to the developing unit; a stopper member configured to be movable between a regulating position for preventing the pressing member from moving from the second position to the first position and a retracting position for permitting the pressing member to move from the second position to the first position, the retracting position being retracted from the regulating position; and an urging member configured to urge the stopper member, wherein, in a case where the developing unit is attached to the apparatus body and the photosensitive member unit is also attached to the apparatus body, the stopper member urged by the urging member is at the retracting position, and wherein, in a case where the developing unit is attached to the apparatus body and the photosensitive member unit is not attached to the apparatus body, the stopper member urged by the urging member is at the regulating position.
 2. The image forming apparatus according to claim 1, wherein, in a case where the developing unit is not attached to the apparatus body and the photosensitive member unit is attached to the apparatus body, the stopper member urged by the urging member is at the retracting position.
 3. The image forming apparatus according to claim 1, wherein, in a case where the developing unit is not attached to the apparatus body and the photosensitive member unit is not attached to the apparatus body, the stopper member urged by the urging member is at the regulating position.
 4. The image forming apparatus according to claim 1, wherein, in a case where the developing unit is not attached to the apparatus body and the photosensitive member unit is not attached to the apparatus body, the stopper member urged by the urging member is at the retracting position.
 5. The image forming apparatus according to claim 1, further comprising a door disposed to be rotatable relative to the apparatus body, and configured to be movable between a closing position for covering the developing unit attached to the apparatus body and an opening position for exposing the developing unit attached to the apparatus body, wherein the pressing member moves from the second position to the first position as the door moves from the opening position to the closing position by the rotation of the door, and wherein the pressing member moves from the first position to the second position as the door moves from the closing position to the opening position by the rotation of the door.
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